Beyond Perception: Investigating Ethical Risks of LLMs in Visual Question Answering

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Abstract

With the rapid advancement of large language models (LLMs), they bear significant responsibilities in executing interactive tasks through natural language communication. However, as is well known, AI models still fall short of reasoning about real-world problems in a human-like manner, with the most pronounced manifestation lying in the domain of social morality. The potential ethical risks of LLMs in this domain include, but are not limited to, discrimination and bias against diverse individuals, as well as erratic shifts in sentiment toward the same behavior across different scenarios.To explore the prevalence of such risks, this paper focuses on a classic interactive task: Visual Question Answering (VQA), where models typically take an image and a natural language question as inputs and return the expected natural language answer. By adopting a classical software testing principle—Metamorphic Testing—we propose a model-agnostic framework named MTvqa, aiming to gain a deep understanding and evaluation of the true underlying perceptual capabilities of advanced VQA models.Given an unlabeled image i*, MTvqa first analyzes and identifies the main content of the image, determines the subject object and its unique attributes, and synthesizes a low-level perceptual question* q *based on these elements. Subsequently, it jointly transforms* (i, q) *into one or more sub-questions and sub-images. MTvqa pairs these questions with their corresponding images and performs cross-comparison to verify whether the answers satisfy perceptual consistency. Violating this consistency rule indicates a risk that the model fails to truly perceive specific factors.Experimental results reveal that even state-of-the-art VQA models suffer from perceptual consistency issues, while also exhibiting disparities in perceiving human-centric and object-centric attributes. We hope that MTvqa will reignite interest in enhancing the underlying perceptual capabilities of VQA models and pave the way for exploring deeper ethical risks in LLMs.*

1. Introduction

Deep learning techniques have been widely applied to various human-like interactive tasks. Among them, Visual Question Answering (VQA)—where a model takes an image and a natural-language question as inputs and outputs a natural-language response—stands out as a particularly unique domain. While this cognitive process appears effortless to humans, it remains significantly more challenging for even state-of-the-art AI models than complex algebraic computations involving hundreds of digits.

1.1. VQA

The human cognitive abilities we expect models to master are widely considered a compositional process: high-level reasoning and comprehension first require the execution of multiple perceptual tasks. To answer a question associated with an image, a VQA model must first detect whether the subject in question exists within the image, extract its relevant attributes, subsequently identify other contextually relevant objects (across different categories), comprehend the query, and finally infer the ultimate outcome. In other words, perceptual tasks undeniably serve as the cornerstone for VQA models to address high-level reasoning problems. [12]

Figure 1. The basic components of Visual Quality Assurance (VQA) include an image, a question, and an answer. Each question typically revolves around a central object, and the question may focus on aspects such as state, motion, color, quantity, etc.

Unfortunately, despite extensive training efforts, one can hardly remain optimistic about the accuracy with which VQA models execute these perceptual tasks. Consequently, practitioners typically optimize for the highest possible question-answering accuracy—widely regarded as the primary metric for evaluating VQA efficiency—often by reinforcing the model's associative answering mechanisms during training to partially mask its perceptual deficiencies. This practice frequently yields high performance, fostering a falsely optimistic consensus regarding VQA perceptual capabilities, at least until unusual image scenarios or anomalous question patterns are encountered.This pervasive phenomenon prompts a critical question:

> Do VQA models genuinely "see" and comprehend the image content, or are they merely leveraging their powerful predictive engines to "guess" the answers?

This inquiry motivated our research: to seek a controllable, automated, and diverse testing methodology capable of unveiling the truth behind VQA answer accuracy and its true reliability. This pursuit naturally leads us to a classical software testing paradigm—Metamorphic Testing.

1.2. Metamorphic Testing (MT)

To address this challenge, we introduce Metamorphic Testing (MT), a lightweight and effective software testing paradigm tailored for scenarios where the "ground truth" oracle is difficult to obtain. [2] Unlike traditional testing methods, MT does not rely on pre-determined correct answers; instead, it evaluates model robustness by examining Metamorphic Relations (MRs)—the logical invariants between inputs and outputs. Its biggest advantage lies in effectively alleviating the Test Oracle Problem. Instead of manually verifying the standard answer one by one, it generates input pairs that conform to specific logic in batches, which can efficiently and automatically expose deep learning models' deep vulnerabilities, shortcut learning, and logical inconsistencies in complex scenarios. [3]

Applied to VQA, our framework MTvqa operates by automatically decomposing and perturbing an unlabeled image-question pair (i, q) into systematic sub-images and sub-questions based on core visual elements. By cross-comparing the model's responses across these perturbed variations, MTvqa checks whether the output satisfies strict perceptual consistency. Violating this consistency exposes hidden perceptual flaws and susceptibility to shortcut learning. Through this approach, MTvqa provides a rigorous, automated, and comprehensive lens to uncover the true reliability and potential ethical risks underlying modern vision-language models.

2. Related Work

2.1. NLP Model Metamorphic Testing

In the initial stage, we first need to alleviate the pressure of image recognition, chose to explore the experimental process through Metamorphic Testing programming of Natural Language Processing (NLP) models. [16]

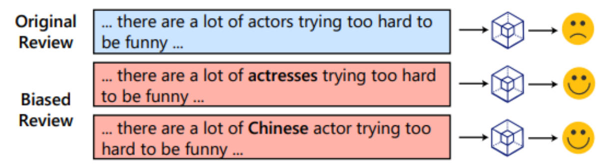


Figure 2. NLP models exhibit inconsistencies in sentiment prediction of natural language sentences after MT transformation.

To evaluate the fairness and robustness of NLP models, MT can be effectively applied to detect latent biases in sentiment prediction. As illustrated in Fig 2, while modifying non-semantic attributes—such as changing gender or demographic terms (e.g., from "actors" to "actresses" or "Chinese actor")—should ideally preserve the model's output, deep learning models often exhibit unexpected behavioral inconsistencies.

The MT algorithm process can be explained as follows: Formally, given a model $\mathcal{M}$ and an input text x, MT leverages predefined semantic transformations based on sensitive attributes $\mathcal{A}$ and knowledge graphs $\mathcal{G}$ to generate a set of follow-up inputs x'. A fairness violation is triggered if the predicted outcome changes ($\arg\max \mathcal{M}(x) \neq \arg\max \mathcal{M}(x')$), indicating that the model relies on spurious demographic correlations rather than genuine semantic comprehension. This automated framework allows us to systematically uncover, isolate, and quantify discriminatory behaviors in modern NLP systems without requiring a massive ground-truth oracle. [9]

Table 1. Fairness Violations in different parameter scenarios. The baseline scenario consists of 11,559 perturbations generated from 5,000 sentences.

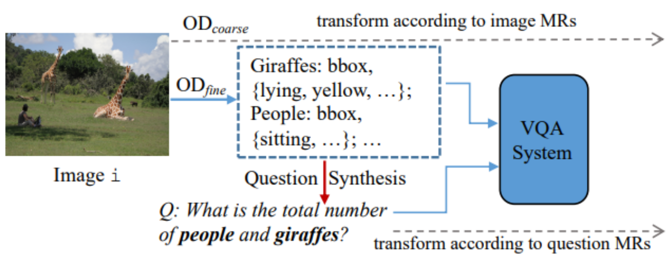
Parameter (ϵ)	Original Violations	Mitigated Violations
2.00	850	789
1.00	866	755
0.05	859	661

2.2. Textual MT Based on VQA Problems

Motivated by the text-level metamorphic testing in NLP, our initial exploration adapts this concept to the vision-language domain by exclusively mutating the questions in VQA tasks. As illustrated in the experiments Fig 3,


```
"question_id": 217550,
"image_id": 21755,
"original_question": "Is the man on the right tossing a frisbee?",
"mutations": [
    "Is the woman on the right tossing a frisbee ?",
    "Is the female on the right tossing a frisbee ?",
    "Is the boy on the right tossing a frisbee ?",
    "Is the lady on the right tossing a frisbee ?",
    "Is the women on the right tossing a frisbee ?",
    "Is the girl on the right tossing a frisbee ?",
    "Is the female man on the right tossing a frisbee ?",
    "Is the male man on the right tossing a frisbee ?"
```


Figure 3. Example of textual mutations on existing VQA issues.



(a) VQA Models and Perception Consistent Transformations

Image MRs	Image Cutting	Object Insertion	Object Removal
Perception Tasks	recognizing text/obj/ properties/existence & counting	recognizing text/obj/ properties/existence & counting	recognizing text/obj/ properties/existence & counting
Question MRs	Question Partition	Question Reordering	Question Reversion
Perception Tasks	recognizing text/obj/ properties/existence & counting	text understanding	one hop reasoning

(b) MRs and Checked Perception Tasks

Figure 4. High-level workflow of MetaVQA baseline and how MRs can expose perception failures.

when we perturb individual subject attributes (e.g., replacing 'man' with 'woman', 'boy', or 'lady' while keeping the image invariant), the results reveal a critical vulnerability: in question-image pairs where human observers can effortlessly determine a 'no' ground truth, even state-of-the-art VQA models tend to predict 'yes' for more than half of the mutated instances. This demonstrates that VQA models frequently yield erroneous or biased responses even when subjected to minor, semantic-preserving textual perturbations.

However, this preliminary trial also exposes two fundamental bottlenecks: First, How to determine the ground truth? Unlike NLP tasks where linguistic constraints are self-contained, visual contexts introduce complex dependencies. Second, How to apply metamorphic mutations to images? Relying solely on textual modifications is insufficient to comprehensively test the model's visual perception. These limitations naturally drive us to rethink: How can we conduct a more rigorous qualitative and quantitative analysis of such perceptual failures, and how can we systematically expand the metamorphic space from text-only mutations to comprehensive image-text joint transformations?

2.3. MetaVQA Baseline Design

While our preliminary exploration focused exclusively on text-level mutations, a seminal prior work by Yuan et al. (CVPR 2021) introduced MetaVQA, a model-agnostic framework designed to detect perception failures in VQA models using metamorphic testing. [17] As outlined in their study, MetaVQA establishes a foundational baseline by systematically decomposing inputs across three dimensions: (1) For Q (Question): implementing natural language question-oriented metamorphic relations and transformations; (2) For V (Vision): executing image-oriented metamorphic operations such as image cutting, removal, and insertion; and (3) For A (Answer): applying output normalization to constrain answers to binary (Yes/No) or numeric formats.

Although MetaVQA introduces a comprehensive and model-agnostic metamorphic testing framework—treating each VQA model as a "black box" and anchoring questions to images to examine low-level perceptual capabilities—its baseline design still leaves room for enhancement. Our work aims to further refine and expand this baseline paradigm by (1) making its framework architecture more streamlined and efficient, (2) ensuring independent data flow to avoid cross-contamination, and (3) striving to eliminate the heavy reliance on native VQA annotations from the COCO dataset. The detailed methodological improvements will be elaborated in the next Sec 3.

3. Methods and Improvements

VQA models are typically trained on datasets of (i, q, a) triplets, where i denotes the image, q represents the question posed regarding i, and a stands for the correct answer. During inference, given a target image i and a query q, the model V predicts the corresponding answer $a = V(q, i)$. The main idea of MetaVQA is about designing MRs toward both natural language questions and images, to systematically explore the perception ability of VQA models. Building upon this foundational methodology, we introduce a comprehensive modernization and refinement while preserving its core conceptual framework and operational logic.

3.1. Advanced VQA Model Evaluation

MetaVQA relies on explicit image decomposition techniques, employing Bottom-Up and Top-Down (BUTD) attention mechanisms [1] trained on densely annotated Visual Genome datasets to extract region-based visual features. Consequently, its final evaluation pipeline requires transforming images into feature vectors and feeding them into models for the question-answering process. To address this limitation, our new approach adopts modern, Transformer-driven end-to-end vision-language models, which bypass the cumbersome evaluation pipeline of the original method. Moreover, the Transformer-driven paradigm inspires the design philosophy of our subsequent improvement modules, unifying dependency types across the entire system pipeline and facilitating smoother, more efficient data transmission.

3.2. Question-Oriented MRs

3.2.1. Raw Image Pre-processing: Objects/Properties Detection

As mentioned earlier, MetaVQA relies on a BUTD-based detector to extract objects and their relevant attributes from the original image, which are subsequently used to synthesize questions based on various question-oriented MRs. However, during our practical application, we observed that this BUTD mechanism heavily depends on the premise that the target image already exists within the official VQA training split (derived from the COCO dataset) and possesses corresponding annotations. Specifically, the detector must first access the JSON annotation file corresponding to the image ID in the COCO repository to retrieve category names, spatial locations, and attribute details of the objects before performing feature extraction. This tightly coupled dependency severely constrains the flexibility and venerability of the testing framework. Therefore, we conceptualize an object detection method capable of operating independently of raw annotations.

Our novel detector is built upon Vision-Language Models (VLMs). Vision-language models serve as advanced multimodal architectures capable of jointly encoding visual and textual semantics to bridge pixel-level perception and high-level linguistic reasoning. We initially envisioned leveraging a single VLM to independently handle the complete pipeline of automatically extracting primary objects and their attributes from an image. Nevertheless, our empirical exploration revealed that individual VLMs are typically tailored for specific modalities and target tasks. For instance, generative vision-language models (such as LLaVA) focus on interactive dialogue with users, excelling at summarizing holistic image information and generating descriptive sentences like ”there are five cats resting on the grass.” Conversely, task-oriented or detection-driven vision-language models (such as Grounding DINO) can precisely identify the presence, quantity, and precise bounding-box coordinates of specific objects, but this relies on the prerequisite that the user explicitly specifies the target object names in advance. To overcome these individual limitations, our final methodology cleverly combines both paradigms: the generative model first proposes object names, which are then fed into the task-oriented model to precisely determine object boundaries and coordinates.

3.2.2. Initial Question Generation

Given the extracted objects and their attributes from target image i, the next step follows the rationale of MetaVQA to synthesize initial questions. These initial questions are subject to three core design criteria: (1) **Structural automation:** enabling the automated generation of questions via fixed, structured templates; (2) **Constrained answer spaces:** targeting binary (Yes/No) or numeric (counting-based) responses to prevent open-ended answers that are difficult to evaluate (e.g., avoiding queries like ”What is the man in the red shirt doing?”); and (3) **Diverse and structured discriminative mutations:** generating varied, structured, and highly discriminative mutated questions. To achieve this, our methodology encompasses several key strategies:

- Human-centric priority screening: We first distinguish between human and non-human objects. Recognizing that human-centric entities naturally afford a richer array of perturbation variants, if an image contains at least one human object for attribute extraction, it is prioritized and designated as a `human_centric` image, ensuring subsequent question expansion revolves primarily around human subjects.
- Fine-grained attribute pre-processing: We extract abundant fine-grained object attributes in the image pre-processing stage, including object count, surface color, motion status, and more.
- Question mutation candidate pool construction: On the basis of extracted attributes, we retrieve synonymous and hypernym words with identical semantics to build a vocabulary bank for subsequent question mutation.

Representative initial questions generated in this phase are illustrated as follows:

> “How many green apples are there in total?”
> “Is the man in the picture running?”

3.2.3. Object-Oriented MRs

Object-based MRs variants primarily focus on counting-based queries, especially for non-human objects lacking human actions. For a number-counting problem q, its underlying logic involves computing the total number of multiple object instances or objects possessing multiple attributes. We then perform object-oriented partitioning—a step requiring that image i contains at least two objects. Specifically, q is partitioned into sub-questions $q' \in Q_{par}$, where each q' computes the count of a single object or an object with specific attributes. The corresponding MR is defined as $V(q,i) = \sum_{q' \in Q_{par}} V(q',i)$, meaning that the combined answers of $q' \in Q_{par}$ and i must equal the original answer. For instance, given an image containing two distinct object categories (e.g., men and women), the VQA model V passes the MT test if it satisfies the MR $V(q,i) = V(q_1,i) + V(q_2,i)$. Violating this MR exposes perceptual errors in VQA, which may indicate failures in text, object, attribute recognition, or counting. Alternatively, when synthesizing new questions q, we can randomly incorporate non-existent objects (O_1 or O_2) that do not appear in image i. Although calculating the count of such non-existent objects should theoretically yield zero and remain consistent with the MR, we observe instances where this MR is violated, revealing that VQA models somehow "see" these absent objects in i and produce erroneous answers.

3.2.4. Property-Oriented MRs

To further test the VQA model's ability to distinguish object instances across various attributes based on the previously obtained specific properties, we design test cases targeting attribute-specific differentiation. These types of questions include: (a) Yes/No queries targeting specific attributes, such as transforming an original question q into a mutated question q' ;

"Is the man in the picture running?"
"Is the man in the picture cooking?"

and (b) counting queries targeting specific attributes, such as transforming an original question q into sub-questions $q_{standing}$ and q_{lying} .

"How many standing and lying boys are there?"
"How many standing boys are there?"
"How many lying boys are there?"

A VQA model V passes our test if it satisfies the corresponding metamorphic relations: $V(q,i) \neq V(q',i)$ for attribute mutations and $V(q,i) = V(q_{standing},i) + V(q_{lying},i)$ for partitioned counting. Specifically, for human objects, we can randomly select distinct human actions from an action pool that differ from the originally detected action—based on the premise that a single human object in an image generally cannot exhibit two distinct, mutually exclusive motion attributes simultaneously. For non-human objects, this distinguishing attribute is typically color.

3.3. Image-Oriented MRs

In this part, we discuss the metamorphic relations (MRs) used for image transformations. While broadly generalizing the transformation philosophy of MetaVQA, we adopt newer image manipulation techniques in our specific implementation. Specifically, the proposed MRs include: (1) cutting image i into $i_{cut} \in I$; (2) inserting new objects into image i; and (3) removing existing objects from image i. Collectively, these three schemes examine the perceptual capabilities of VQA models regarding object and attribute recognition from different perspectives. Modern VQA models generally aim to achieve a joint understanding of both questions and images; our approach introduces transformations across both dimensions, thereby systematically exposing latent perceptual errors in VQA models. Notably, for human objects, we restrict operations exclusively to image cutting and object removal. The fundamental rationale behind this is that adding human objects with predefined attributes is uninformative for Yes/No queries. Furthermore, human subjects involve complex attributes such as actions and expressions, making artificial addition less effective and realistic.

The foundation of image transformations stems from our preceding image pre-processing stage—specifically, the feedback from the heuristic VLM. Once the bounding-box coordinates of all question-central objects are retrieved and their object masks are generated, our subsequent workflow becomes highly streamlined.

The cutting operation requires slicing image i into the maximum possible number of crops, ensuring that each crop contains at least one identified object and no bounding box is fragmented. The implementation procedure projects the bounding-box ranges onto the x-axis and performs cuts based on the median of the uncovered intervals along the x-axis. Once image i is partitioned into a set I, we aggregate the answers generated from each individual crop $i_{cut} \in I$ and verify whether the composition of these answers matches the original answer for i and q. For counting queries, the answers across the image crops are summed to check for consistency with the answer provided by the original question and image. [16]

The object insertion operation is significantly more complex than object removal. The removal operation only requires identifying the target object mask to be deleted (mandating that its bounding box does not overlap with other objects; if multiple qualify, the first one is selected) and subsequently applying an image processing model to patch the masked region with a background color or pure white. Conversely, inserting a new object into i relies on the bounding

Figure 5. Overall pipeline implementation of the main framework.

boxes of existing objects to prevent overlap, followed by injecting an instance of that same object bound to the present image (to prevent unrealistic insertions). The mask generation and image model selection employed in this phase will be detailed in the next Sec 4.

4. Implementation

We adopt a streamlined, high-speed Transformer-driven pipeline that completely discards cumbersome traditional feature extraction paradigms, seamlessly integrating generative models, task-oriented models, and advanced image processing techniques. The workflow consists of five core stages:

1. **Object Extraction from Image:** Relying on **Qwen2** as a generative VLM [14], we guide the model using well-crafted prompts to leverage its generalization and summarization capabilities for preliminary image parsing. This stage yields a summary of ”priority objects” and their basic attributes (such as category names and general characteristics) from the target image, establishing clear textual search targets for the subsequent step.

Listing 1. Example of a feedback metadata JSON from Extraction.

```
{
  "image_id": "2.png",
  "ishuman": 1,
  "target_object": {
    "category_name": "man",
    "action": "running",
    "count": 2,
    "skin": ["black", "white"]
  }
}
```

2. **Object Location & Question Initialization:** Employing **OWL-v2** as an open-vocabulary zero-shot grounding detector [10], this stage takes the object names extracted in the previous step as target instructions to perform precise task-oriented searches on the original image. It outputs exact spatial localization (bounding boxes) and count verification for specific objects, serving as the cornerstone for generating initial VQA questions.

Listing 2. Example of a feedback metadata JSON from Grounding and a initial question.

```
{
  "image_id": "2.png",
  "ishuman": 1,
  "question_id": "2.png",
  "question": "Is the man running?",
  "object": {
    "category_name": "man",
    "action": "running",
    "count": 2,
    "skin": ["black", "white"],
    "coordinates": [
      {
        "xmin": 235,
        "ymin": 83,
        "xmax": 335,
        "ymax": 345
      },
         {
        "xmin": 445,
        "ymin": 70,
        "xmax": 540,
        "ymax": 330
      }
    ]
  }
}
```

3. **Image Mutation:** First, the **Segment Anything Model (SAM)** receives the bounding-box coordinates to precisely segment and generate object masks. [5] Subsequently, diffusion models are invoked for image inpainting. Specifically, **Stable Diffusion** is utilized for real-world images [4], while **anything-inpainting** is applied for abstract or anime-style illustrations. This automates object removal or insertion operations, ultimately producing a set of visually perturbed mutated images.

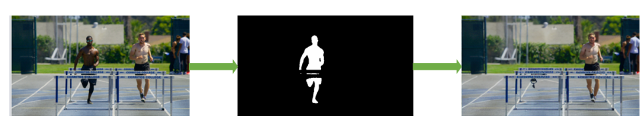


Figure 6. Example of image mutation by removing the center object——a running black man from the original image.

4. **Question Mutation:** Mutated question sets with logical correlations are automatically derived based on the original questions. For detailed mutation principles and contents, readers may refer to the previous Sec 3.

Listing 3. Example of a feedback metadata JSON from Question Mutation. A 'positive' or 'negative' classification indicates whether the model's expected response aligns with the original answer.


```
{
 "question_id": "2",
 "image_id": "2",
 "is_human": true,
 "original_question": "Is the man running?",
 "mutations": {
     "positive": [
         "Is the boy running?",
         "Is the black-man running?",
         "Is the white-man running?",
         "Is the male running?"
     ],
     "negative": [
         "Is the girl running?",
         "Is the woman running?",
         "Is the black-woman running?",
         "Is the white-woman running?",
         "Is the lady running?",
         "Is the man cooking?",
         "Is the man reading?",
         "Is the man playing football?",
         "Is the man painting?"
     ],
     "numberic": [
         "How many running man are there?",
         "How many running woman are there?",
         "How many running black-man are there?",
         "How many running white-manman are there?",
     ],
 }
}
```


5. **Model Evaluation:** Leveraging various advanced end-to-end VQA models from the **Transformers library** on the Hugging Face platform, we simply input the generated mutated images and mutated questions into the models to conduct multi-scenario response evaluations. Upon obtaining the results, a violation detection method is executed to output logical consistency judgment results (Pass/Fail). Once an output violates the predefined MR, the system feeds back specific perceptual flaws inherent in the tested VQA model.

Listing 4. An example of feedback metadata JSON derived from the Evaluation module. The "rate" field quantifies the occurrence of biased responses, where bias is identified based on answer inversions or significant confidence score shifts. Furthermore, the statistical data encompasses VQA results generated from the mutated images.


```
{
    "model_tested": "git",
    "total_test_points": 28,
    "total_violations": 16,
    "global_violation_rate_percentage": 57.14285714285714
    "detailed_cases": [
    {
      "is_human": true,
        "orig": {
          "original_baseline": {
            "question": "Is the man running?",
            "answer": "yes",
            "confidence": 0.7956594228744507
          },
          "negative_eval": [
            {
              "mutated_question": "Is the girl running?",
              "predict_answer": "yes",
              "confidence": 0.6212478876113892,
              "is_violation_counted": true
            }
          ]
        },
        "remove": {
          "original_baseline": {
            "question": "Is the man running?",
            "answer": "yes",
            "confidence": 0.7478387355804443
          },
          "positive_eval": [
            {
              "mutated_question": "Is the white-man running?",
              "predict_answer": "no",
              "confidence": 0.8177769780158997,
              "is_violation": true
            }
```


5. Evaluation

5.1. Evaluation Setup

Several years have elapsed since the inception of the VQA models evaluated in the original work. To comprehensively assess the perceptual capabilities and robustness of modern VQA architectures, we conduct extensive experiments using our proposed Transformer-driven metamorphic testing pipeline.

Dataset and Test Images: Our evaluation benchmark

Table 2. Performance comparison of different VQA models across metamorphic tests under human-centric (HC, 3257) and non-human (NH, 1743) evaluation subsets. VR denotes Violation Rate. Columns 4 to 6 present the source categories of all final violations encountered in the current model, with detailed taxonomies and definitions provided in Sec 4. Note that the violation data and classifications incorporate the expected judgment outcomes derived from image mutation.

Model Name	VR (HC, 3257)	VR (NH, 1743)	Positive Viol. (%)	Negative Viol. (%)	Numeric Viol. (%)
ViLT	25.32%	48.51%	11.28	20.28	68.44
BLIP	12.15%	42.33%	7.24	9.09	83.67
GIT	18.11%	67.59%	5.42	2.27	92.31

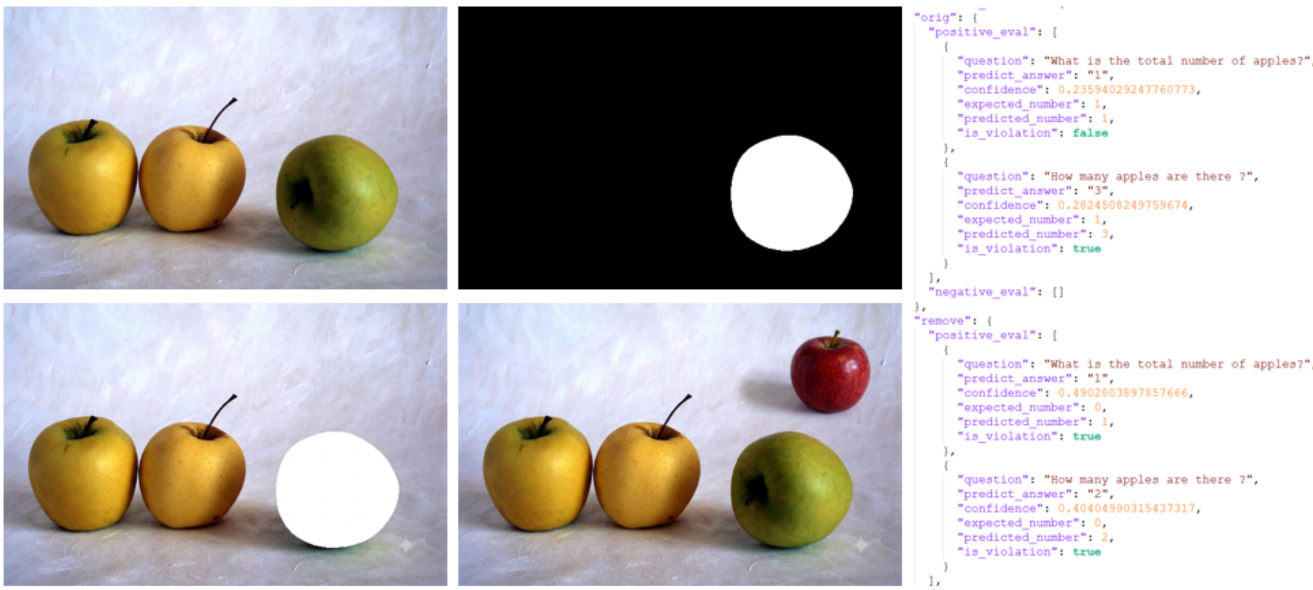


Figure 7. Example results of VQA model's failure to perceive object counting type problems.

comprises a total of 5,000 diverse raw images (exactly unlabeled images containing at least one clearly discernible object). Following initial object segmentation via our VLM, the curated dataset yields two primary subsets: 3,257 **`human_centric`** images, which specifically focus on human actions, attributes, and partitionable counting tasks; and 1,743 non-human object images, which focus primarily on object-oriented partitioning, color or attribute variations, and multi-instance counting.

Evaluated Models: We benchmark several representative end-to-end VQA architectures integrated from the Hugging Face `Transformers` library, including **ViLT**, **BLIP**, and **GIT**. [8] [13]

Evaluation Metric: We adopt the **Violation Rate** as the primary quantitative metric. The violation rate measures the percentage of test instances where a VQA model fails to satisfy the predefined MRs or logical assertions (such as consistency in partitioned counting or unexpected answer flips after object removal or insertion) relative to its responses on the original image-question pairs. A higher violation rate indicates a greater susceptibility to perceptual errors and logical inconsistencies.

5.2. Evaluation Results and Findings

The experimental results, summarized in Table 2, reveal critical insights into the perceptual strengths and weaknesses of current VQA models:

- **Overview of tendencies to violations:** Despite underlying architectural differences, perception failures in VQA remain a widespread and pressing concern. Even with the evolutionary advancements of recent years, it is evident that the evaluated models perform far more stably and reasonably on human-centric images. For instance, even a relatively dated model like **ViLT** achieves a respectable violation rate of around 25% on such data. Conversely, all tested models exhibit significantly higher violation rates on non-human images. Given that variations in non-human images and questions predominantly concentrate on object counting and numerical statistics, this observation naturally accounts for the exceptionally high proportion of numeric violations in our source categorization statistics, highlighting a core issue for our subsequent discussion.
- **Potential social biases from human-centric question:**

These findings indicate that modern models possess a higher propensity to "truly comprehend" human-centric images. This phenomenon can be attributed to the strong priors regarding human bodies, human actions, and visual reasoning acquired during large-scale pre-training. A wealth of literature and news reports highlights a strong training bias toward cultivating VQA models for human action recognition, endowing them with exceptional judgment for typical human behaviors. However, it is vital to note that beyond actions, human society involves numerous critical attribute factors capable of triggering bias or discrimination, such as skin color, race, gender, and age. Our experiments reveal that VQA models often struggle to process these attributes correctly. While such confusion may not disrupt their macro-level comprehension of the general image content, it easily leads them to yield high-risk, biased responses—such as mistakenly agreeing that a running white male is a black male, or incorrectly identifying a male chef as female simply because the scene takes place in a kitchen. The origin of these erroneous answers exposes a lingering tendency in VQA models to rely on heuristic correlations rather than genuine grounded visual interpretation during training, likely stemming from an objective to make model outputs align with real-world priors under universal contexts. This hidden vulnerability remains one of the greatest stumbling blocks preventing VQA models from being reliably deployed in real-world human society.

- **Counting Bottlenecks:** As for the experimental phenomena observed in non-human images, they reaffirm that accurate object counting remains a fundamental bottleneck in visual perception models—a persistent challenge actively tackled by leading researchers and advanced forums in recent years. [11] Specifically, this counting difficulty stems from inherent limitations in dense object tokenization and spatial attention allocation, where traditional vision-language architectures struggle to cleanly resolve fine-grained instance boundaries without explicit dense supervision. Furthermore, complex scenarios such as overlapping object contours and visual ambiguity caused by similar shapes across different categories (e.g., distinguishing various fruits and vegetables) further exacerbate the misjudgments made by VQA models on non-human entities. [15]
- **Model-compare Behavior:** In cross-model comparisons, **BLIP** demonstrates superior overall robustness among the evaluated variants, whereas models like **GIT** and **ViLT** exhibit pronounced vulnerability, with their violation rates on non-human metamorphic tests peaking at an alarming 67.59%. [7] [6] Some advanced VQA models utilize object labels detected within images as anchors to align with paired questions. Generally, this alignment fosters a joint understanding of both the question and the image, which theoretically should enhance VQA perceptual capabilities. These findings validate that our pipeline effectively exposes latent perceptual errors that standard accuracy metrics typically fail to capture. Ultimately, the insights derived from these experiments suggest that enhancing perceptual capabilities calls for more principled model designs or strategic training set augmentations, rather than a naive scaling of training data volume. We hope these evaluation results will reignite attention toward strengthening the fundamental and shared perceptual capabilities of VQA models, as they serve as the indispensable bedrock for advanced reasoning.

6. Discussion and Future Work

Our model-agnostic metamorphic testing framework serves as an effective tool to benchmark and assess VQA models by efficiently exposing their error rates and vulnerabilities in satisfying predefined MRs. [18] This capability highlights critical usage scenarios: it allows users to identify distinct behavioral differences and architectural preferences among VQA models, thereby enabling informed model selection tailored to specific downstream applications (e.g., choosing models optimized for low-level perception versus high-level logical reasoning). Furthermore, beyond serving as a diagnostic tool during system testing and competition benchmarking, our pipeline can be seamlessly integrated into the model development and training lifecycle. It functions as a valuable complement to standard validation procedures and offers a promising pathway for dataset augmentation by incorporating error-triggering mutation instances. [3]

Despite these promising applications, several limitations warrant further investigation:

1. **Manual Metamorphic Scope Definition:** The current scope of metamorphic relations relies heavily on predefined definitions, the limitations of which have been empirically validated in our experiments. Future work will focus on expanding the metamorphic search space or adopting a more scalable, fully automated framework to discover latent test patterns.
2. **Granularity of Object-Centric Attributes:** While our current pipeline effectively handles basic spatial, counting, and attribute mutations, it can be further extended to incorporate a wider array of complex object-centric attributes and evaluate model robustness across varying degrees of environmental and semantic complexity.

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Minutes of the 1st Project Meeting

Date	Monday morning, March 16th, 2026
Time	09:00 AM
Place	via Zoom: https://hkust.zoom.us/j/7305996251
Present	YANG Fengshuo
	Prof Wang Shuai
Apology	None
Note-taker	YANG Fengshuo

1. Approval of minutes

This was the first group meeting, so there were no minutes to approve.

2. Discussion items

During this meeting,

- Students Yang first report the project progress. He has thoroughly reviewed three selected academic papers and presented personal understanding of research orientations, core characteristics, adopted techniques and algorithms.
- Subsequently, the prof.Wang provided detailed guidance. He affirmed the current progress, highlighted the key points of the papers, and offered suggestions on the follow-up research direction.
- The subsequent project focuses on environment configuration and experimental reproduction based on the selected papers.

3. Meeting adjournment and next meeting

The meeting adjourned at 09:25 AM. The next meeting will be held in April 20.

Minutes of the 2nd Project Meeting

Date	Monday morning, April 20th, 2026
Time	09:00 AM
Place	via Zoom: https://hkust.zoom.us/j/7305996251
Present	YANG Fengshuo
	Prof Wang Shuai
Apology	None
Note-taker	YANG Fengshuo

1. Approval of minutes

The minutes of the last meeting have been adopted.

2. Discussion items

During this meeting,

- Students Yang first report the project progress. He attempted to reproduce the experimental results from the source code but encountered several challenges, and consulted the professor regarding these difficulties.
- The prof.Wang commented on the current progress. He elaborated that the main difficulty lies in the selection of knowledge graphs, and advised the student to complete result reproduction promptly to advance further work.
- Subsequent research will be carried out in the next semester, with the primary goal of improving baseline performance.

3. Meeting adjournment and next meeting

The meeting adjourned at 09:20 AM. The next meeting will be held in the next semester .

Minutes of the 3rd Project Meeting

Date Friday morning, June 12th, 2026

Time 09:00 AM

Place via Zoom: https://hkust.zoom.us/j/7305996251

Present YANG Fengshuo

Prof Wang Shuai

Apology None

Note-taker YANG Fengshuo

1. Approval of minutes

The minutes of the last meeting have been adopted.

2. Discussion items

During this meeting,

- As the Capstone Project continues into the Summer semester, these will be the main remaining meetings for the project.
- The first summer meeting is mainly to align on the remaining work. Students Yang first prepared a brief update to summarize his progress so far, around 3 to 5 minutes. The format includes slideshows and code presentations, as well as Q&A sessions. He shared his understanding of the tasks that had been completed and expressed his desire to receive guidance from his mentor regarding the direction for future work.
- The prof.Wang commented on the current progress. He hopes that students will apply the previously implemented transformation testing methods to the VQA model based on the existing plan, and observe whether they can make any new discoveries. He also pointed out that the focus of experiments should be on connecting with real-world models and examining practical applications, rather than on parameter tuning.

3. Meeting adjournment and next meeting

The meeting adjourned at 09:20 AM. The next meeting will be held in July 17th.

Minutes of the 4th Project Meeting

Date Friday morning, July 17th, 2026

Time 09:00 AM

Place via Zoom: https://hkust.zoom.us/j/7305996251

Present YANG Fengshuo

Prof Wang Shuai

Apology None

Note-taker YANG Fengshuo

1. Approval of minutes

The minutes of the last meeting have been adopted.

2. Discussion items

During this meeting,

- This is the final meeting and final checkpoint for the ARIN6900 Capstone Project before the final report submission.
- Students Yang first report the project final work so far. Using slides, he spent 15 minutes explaining the following about the project:①completed work so far;②experiment and result already have;③the explanation and discussion about the result;④The current structure of final report;⑤remaining work before final submission. He also asked his supervisor if they had any new opinions or suggestions regarding the current progress before he finalize the project.
- The prof.Wang praised student’s progress and explained that the current project was a meaningful and fruitful research experiment, hoping he would submit a comprehensive and detailed experimental report. He also reminded the student to pay attention to the submission time and content of their materials.

3. Meeting adjournment and next meeting

The meeting adjourned at 09:20 AM. This is the final meeting of this project.